Introduction to Reliability

- It is the probability that an item, a product, piece of equipment, or system will perform its intended function for a stated period of time under specified operating conditions.
- Simply, reliability means how long an item (such as a machine) will perform its intended function without a breakdown.
- Reliability is performance over time, probability that something will work when you want it to.

Important elements

- Probability
- Performance
- Time
- Operating conditions.

Why Reliability

- The expectation of the customer about certain product are Reliability, Availability, Maintainability, Safety and Quality.
- Companies who control the Reliability of their products can only survive in the business in future as today's consumer is more intelligent and product aware.
- The complexity of products is increasing everyday and thus challenge to Reliability Engineering is also increasing.
- Products are also being advertised by their Reliability Ratings.

Basic Reliability Terms

- Failure - A failure is an event when an item is not available to perform its function at specified conditions when scheduled or is not capable of performing functions to specification.
- Failure Rate - The number of failures per unit of gross operating period in terms of time, events, cycles.

Basic Reliability Terms

- MTTF - Mean Time To Failure - The average time to failure occurrence. The number of items and their operating time divided by the total number of failures. For Repairable Items and Non-repairable Items
- Hazard - The potential to cause harm. Harm including ill health and injury, damage to property, plant, products or the environment, production losses or increased liabilities.
- Risk - The likelihood that a specified undesired event will occur due to the realisation of a hazard by, or during work activities or by the products and services created by work activities.

Basic Reliability Terms

- Maintainability - A characteristic of design, installation and operation, usually expressed as the probability that an item can be retained in, or restored to, specified operable condition within a specified interval of time when maintenance is performed in accordance with prescribed procedures.
- MTTR - Mean Time To Repair - The average time to restore the item to specified conditions.
- Maintenance Load - The repair time per operating time for an item.

Basic Reliability Terms

- Availability - A measure of the time that a system is actually operating versus the time that the system was planned to operate. It is the probability that the system is operational at any random time t.
- Supportability - The ability of a service supplier to maintain the Plant inbuilt reliability and to perform scheduled and unscheduled maintenance according to the Plant inbuilt maintainability with minimum costs.

- As Reliability Engineering is concerned with analyzing failures and providing feedback to design and production to prevent future failures.
- Reliability engineers usually speaks of
- **Failures Causes**
- **Failure Modes**
- **Failure Mechanisms**

*A **failure** is an event at which the system stops to fulfill its specified function.*

- **Reliability measurement** is based on the failure rate

$$Failure\ rate = \frac{Items\,Failed}{Total\ Operating\ Time}$$

- Some products (Non-repairable) are scrapped when they fail e.g. bulb
- Other products (Repairable) are repaired e.g. washing machine.

Reliability function

- This function gives the probability of an item operating for a certain amount of time without failure. As such, the reliability function is a function of time, in that every reliability value has an associated time value. In other words, one must specify a time value with the desired reliability value, *i.e.* 95% reliability at 100 hours.

Reliability function

- The reliability function of a system or a piece of equipment at time t is defined by $R(t) = P(T \geq t)$.
- Where T, the failure time, is a random variable with a known distribution.
- T is also called : Life length or time to failure of a component.
- T is continuous random variable with some probability density function $f(t)$.

Reliability function

- $R(t) = P(T \geq t) = \int_t^{\infty} f(t)dt = 1 - P(T < t)$.
- $R(t) = 1 - F(t)$; F(t) =cdf of T.
- $f(t) = -\frac{d\left(R(t)\right)}{dt} = -R'(t)$.
- Remark:
- Since $R(t) = 1 - F(t)$, hence $R(0) = 1$ and $R(\infty) = 0$.

Hazard function

Instantaneous rate of failure function

- Different hazard functions are modeled with different distribution models.
- $\lambda(t) = \frac{f(t)}{R(t)}$.
- Note : $\lambda(t) = -\frac{R'(t)}{R(t)}$
- Thus $R(t) = e^{-\int_0^t \lambda(t)dt}$
- Also $f(t) = \lambda(t)e^{-\int_0^t \lambda(t)dt}$

$\frac{R'}{R} = \lambda(t)$

$\ln R = \int_0^t \lambda(t)\,dt$

$R = e^{\int \cdots}$

Mean time to failure

- Mean time to failure (MTTF):
- $MTTF = E(T) = \int_0^\infty tf(t)dt = \int_0^\infty R(t)dt$.
- Variance of T :
- $Var(T) = E(T^2) - E(T)^2 = \int_0^\infty t^2 f(t)dt - (MTTF)^2$

Conditional Reliability

- Conditional reliability is defined as the probability that a component or system will operate without failure for a mission time, *t*, given that it has already survived to a given time, T_0.
- $R\left(\frac{t}{T_0}\right) = P\left(\frac{T > T_0 + t}{T > T_0}\right) = \frac{R(T_0 + t)}{R(T_0)} = e^{-\int_{T_0}^{T_0+t} \lambda(t)dt}$.

Exponential distribution

- If the time to failure t follows exponential distribution,

$\frac{1}{\beta} e^{-\frac{1}{\beta}x}$

- $f(t) = \lambda e^{-\lambda t}, t > 0.$
- $R(t) = \int_t^\infty f(t)dt = e^{-\lambda t}.$
- $\lambda(t) = \frac{f(t)}{R(t)} = \lambda.$
- This means that when the failure rate follows exponential with the parameter λ, the failure rate at any time is a constant (= λ).
- This distribution is also called as constant failure rate distribution.

Exponential distribution

- MTTF = $\mathrm{E(T)} = \frac{1}{\lambda}$.
- $Var(T) = \frac{1}{\lambda^2}$.
- $R\left(\frac{t}{T_0}\right) = e^{-\lambda t}$.
- This means that the time to failure of a component is not dependent on how long the component has been functioning.

Normal distribution

If the time to failure T follows a normal distribution $N(\mu, \sigma)$ its pdf is given by

$$f(t) =, \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \qquad t > 0.$$

$$MTTF = E(T) = \mu.$$

$$Var(T) = \sigma_T^2 = \sigma^2.$$

Problem 1

The density function of the time failures in years of the gizmos manufactured by a certain company is given by

$f(t) = \frac{200}{(t+10)^3}$ where $t \geq 0$.

i) Derive the reliability function and determine the reliability for the first year of operation.

ii) Compute MTTF.

iii) What is the design life for reliability 0.95?

Solⁿ: i) The reliability function $R(t) = \int_t^\infty f(t)dt = \int_t^\infty \frac{200}{(t+10)^3}dt = \frac{-200}{2(t+10)^2}\Big|_t^\infty = \frac{100}{(t+10)^2}$.

The reliability for first year $= R(1) = \frac{100}{(1+10)^2} = \frac{100}{121} = 0.826$.

ii) $MTTF = E(T) = \int_0^\infty R(t)dt = \int_0^\infty \frac{100}{(t+10)^2}dt = \frac{-100}{t+10}\Big|_0^\infty = \frac{100}{10} = 10$ years.

iii) Let t be the design life for reliability 0.95. Then

$R(t) = 0.95 \Rightarrow \frac{100}{(t+10)^2} = 0.95 = \frac{95}{100} \Rightarrow (t+10)^2 = \frac{100 \times 100}{95} \Rightarrow t+10 = \frac{100}{\sqrt{95}}$

$\Rightarrow t = \frac{100}{\sqrt{95}} - 10 = 0.259$ year $= 0.259 \times 365$ days $= 95$ days.

Problem 2

The time to failure in operating hours of a critical solid-state power unit has the hazard rate function $\lambda(t) = 0.003\left(\frac{t}{500}\right)^{0.5}$, for $t \geq 0$.

(a) What is the reliability if the power unit must operate continuously for 50 hours?

(b) Determine the design life if a reliability of 0.90 is desired.

(c) Compute the mean time to failure.

(d) Given that the unit has operated for 50 hours, what is the probability that will survive a second 50 hours of operation?

Sol^n: a) Given $\lambda(t) = 0.003\left(\frac{t}{500}\right)^{0.5} = \frac{0.003}{\sqrt{500}} t^{\frac{1}{2}}$. Let $a = \frac{0.003}{\sqrt{500}}$. Then $R(t) = e^{-\int_0^t \lambda(t)dt} = e^{-\int_0^t a t^{1/2} dt} = e^{-\frac{2}{3}at^{3/2}}$. Then $R(50) = e^{-\frac{2a}{3}(50)^{3/2}} = 0.9689$.

b) Let t be the design life, then $R(t) = 0.90 \Rightarrow e^{-\frac{2}{3}at^{2/3}} = 0.9 \Rightarrow -\frac{2}{3}at^{3/2} = \ln 0.9 \Rightarrow t^{3/2} = -\frac{3}{2a}\ln 0.9 = 1177.9664 \Rightarrow t = 111.54$ hours.

c) Mean time to failure (MTTF) $= E(T) = \int_0^\infty R(t)dt = \int_0^\infty e^{-\frac{2a}{3}t^{3/2}} dt$.

Let $\frac{2a}{3}t^{3/2} = x \Rightarrow \frac{2a}{3}\cdot\frac{3}{2} t^{1/2}dt = dx \Rightarrow dt = \frac{1}{at^{1/2}}dx$

$\therefore t = \left(\frac{3x}{2a}\right)^{2/3} \Rightarrow t^{1/2} = \left(\frac{3x}{2a}\right)^{\frac{1}{3}}$

$= \frac{1}{a}\cdot\frac{1}{\left(\frac{3x}{2a}\right)^{1/3}}dx$

$= \frac{1}{a^{2/3}}\cdot\frac{1}{(1.5)^{1/3}} x^{-1/3}dx$

$= \int_0^\infty e^{-x}\cdot\frac{1}{a^{2/3}}\cdot\frac{1}{(1.5)^{1/3}}\cdot x^{-1/3}dx$

$= \frac{1}{a^{2/3}}\cdot\frac{1}{(1.5)^{1/3}}\cdot\Gamma(2/3) = 451.37$ hours $(\because \Gamma(2/3) = 1.354)$

d) To find $R\left(\frac{t}{T_0}\right) = R\left(\frac{50}{50}\right) = \frac{R(50+50)}{R(50)} = \frac{R(100)}{R(50)} = \frac{e^{-\frac{2a}{3}(100)^{3/2}}}{e^{-\frac{2a}{3}(50)^{3/2}}} = e^{-\frac{2a}{3}\left[(100)^{3/2} - (50)^{3/2}\right]} = 0.9438.$

Problem 3

The reliability of a turbine blade is given by $R(t) = \left(1 - \frac{t}{t_0}\right)^2, 0 \leq t \leq t_0$, where t_0 is the maximum life of the table.

(*i*). Show that the blades are experiencing wear out.(i.e. show that hazard rate is increasing over time).

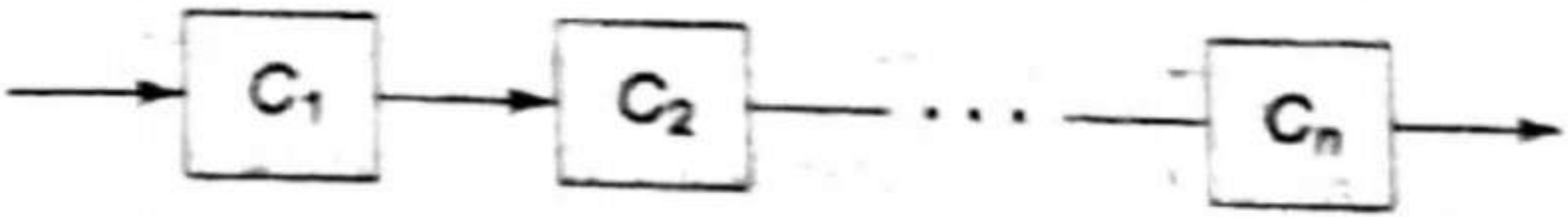
(*ii*). Compute MTTF as a function of the maximum life.

(*iii*). If the maximum life is 2000 operating hours, then compute the design life for a reliability of 0.90.

Problem 4

A manufacturer determines that, on the average, a television set is used 1.8 hours per day. A one-year warranty is offered on the picture tube having a MTTF of 2000 hours. If the distribution is exponential, what percentage of the tubes will fail during the warranty period?

System Reliability
Series or non redundant configuration

- Series(non redundant) configuration is one which the components of the system are connected in series.
- 

Series or non redundant configuration

- In a series configuration all the components must function for the system to function. That is the failure of any component affects the failure of the system.
- Let there are n components (Say C_i , $1 \leq i \leq n$) in the system S which are connected in series. Let $R_i(t)$ be the reliability of the component C_i. Then the reliability of the system S, $R_s(t)$ is obtained as
- $R_s(t) = P(C_1 \cap C_2 \cap \cdots \cap C_n)$

Series or non redundant configuration

- $R_s(t) = P(C_1) * P(C_2) * \cdots * P(C_n)$ $(\because C_i\text{'s are independent})$
- Since $R_i(t) = P(C_i). = R_i$
- $R_s(t) = R_1(t) * R_2(t) * \cdots * R_n(t)$

If C_i's are identical, then $P(C_1) = P(C_2) = \cdots = P(C_n) = R$ (say).

Then $R_s(t) = R.R \ldots R = R^n$.

Parallel or redundant configuration

- Parallel(redundant) configuration is one which the components of the system are connected parallel.
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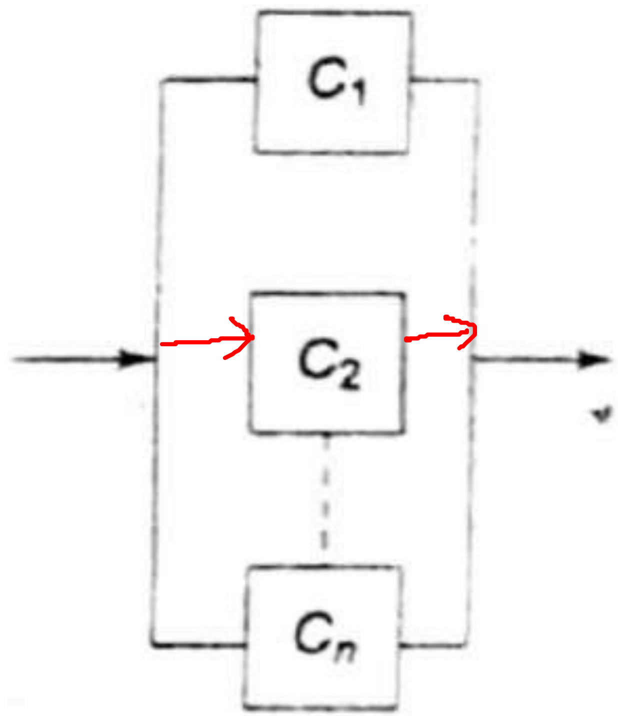


Parallel or redundant configuration

- In a parallel configuration all the components must fail for the system to fail. That is if one or more component function then the system continue to function.
- Let there are n components (Say C_i , $1 \leq i \leq n$) in the system S which are connected in parallel. Let $R_i(t)$ be the reliability of the component C_i. Then the reliability of the system S, $R_s(t)$ is obtained as
- $R_s(t) = P(C_1 \cup C_2 \cup \cdots \cup C_n)$

Parallel or redundant configuration

- $R_s(t) = P(C_1) + P(C_2) + \cdots + P(C_n) - P(C_1 \cap C_2) - \cdots$

If C_1 and C_2 are independent, then $P(C_1 \cap C_2) = P(C_1)\,P(C_2)$.

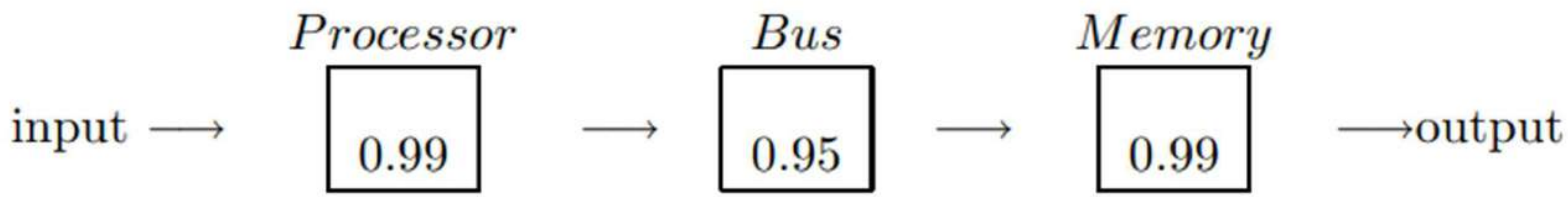
$$\therefore\ P(C_1 \cup C_2) = P(C_1) + P(C_2) - P(C_1 \cap C_2)$$
$$= R_1 + R_2 - R_1 R_2$$
$$= 1 - (1-R_1)(1-R_2).$$

Then, for n independent events:

- Since $R_i(t) = P(C_i) = R_i$.
- $R_s(t) = 1 - (1 - R_1)(1 - R_2) \ldots (1 - R_n)$.

Example 1: A simple computer consists of a processor, a bus and a memory. The computer will work only if all three are functioning correctly. The probability that the processor is functioning is 0.99, that the bus is functioning 0.95, and that the memory is functioning is 0.99 .

Answer:



The probability that the computer will work is:

$$Rel = .99 \times .95 \times .99 = 0.893475$$

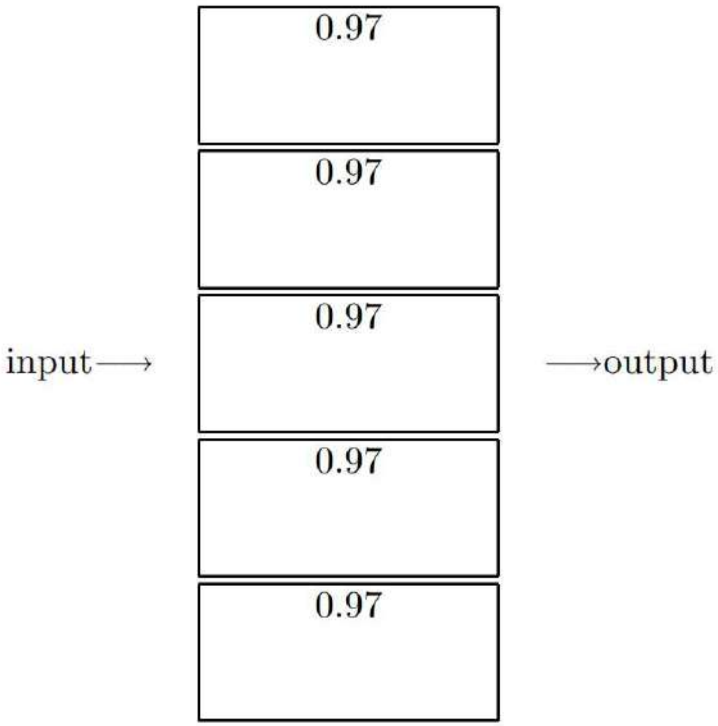
So even though all the components have above 95% or more reliability, the overall Reliability of the computer is less that 90%.

Example 2: An electronic product contains 100 integrated circuits. The probability that any integrated circuit is defective is .001 and the integrated circuits are independent. The product operates only if all the integrated circuits are operational. What is the probability that the product is operational?

Example 3: A system consists of 5 components in parallel. If each component has a reliability of 0.97, what is the overall reliability of the system?

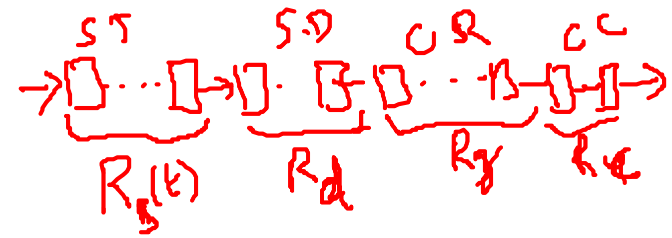
$$R_s(t) = 1 - (1-R_1)(1-R_2)\cdots(1-R_n)$$
$$= 1 - (1-R)^n \qquad \because R = R_1 = R_2 = \cdots = R_n.$$

$$Rel = 1 - (0.03)^5 \approx 1.00000$$



Example 4

- An electronic circuit consists of 5 silicon transistors, 3 silicon diodes, 10 composition resisters and 2 ceramic capacitors are connected in series configuration. The hourly failure rate of each component is given below



- Silicon transistors (5) : $\lambda_s = 4 * 10^{-5}$
- Silicon diodes (3) : $\lambda_d = 3 * 10^{-5}$
- Composition resistors (10) : $\lambda_r = 2 * 10^{-4}$
- Ceramic capacitors (2) : $\lambda_c = 2 * 10^{-4}$
- Calculate the reliability for 10 hours.

each com~ exponential.

Sol^n: Given that the failure rate of each component are $\lambda_s = 4\times10^{-5}$, $\lambda_d = 3\times10^{-5}$, $\lambda_r = 2\times10^{-4}$, $\lambda_c = 2\times10^{-4}$

then reliability $= e^{-\int_0^t \lambda\, dt} = e^{-\lambda t}$ since rates are constants.

Now reliability of 5 Silicon transistors $R_{ST}(t) = (e^{-\lambda_s t})^5 = e^{-5\lambda_s t}$

" " 3 " diodes $R_d(t) = (e^{-\lambda_d t})^3 = e^{-3\lambda_d t}$

" " 10 Composition resistors $R_r(t) = (e^{-\lambda_r t})^{10} = e^{-10\lambda_r t}$

" " 2 Ceramic capacitors: $R_c(t) = (e^{-\lambda_c t})^2 = e^{-2\lambda_c t}$

Hence the system reliability $R_s(t) = R_{ST}(t)\cdot R_d(t)\cdot R_r(t)\, R_c(t) = e^{-(5\lambda_s + 3\lambda_d + 10\lambda_r + 2\lambda_c)t}$

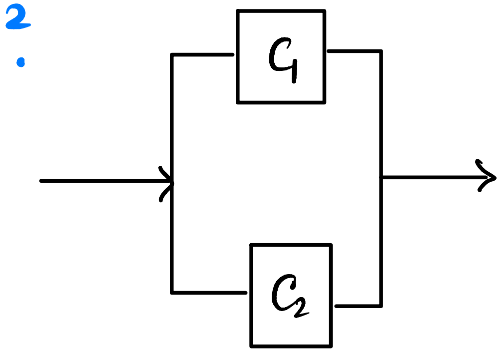
Therefore, the reliability for 10 hours $= R_s(10) = e^{-0.0269}$

$= 0.9738.$

Example 5

- Two parallel, identical and independent component have constant failure rate. It is desired that $R_s(1000) = 0.95$. Find the MTTF for the components and system.

Sol^n: Let λ be the constant failure rate. Then reliability of each component $R(t) = e^{-\int_0^t \lambda dt} = e^{-\lambda t}$.



Then the reliability of the system $R_s(t) = 1-(1-R(t))^2 = 1-(1-e^{-\lambda t})^2$.

Given that $R_s(1000) = 0.95 \Rightarrow 1-(1-e^{-\lambda t})^2 = 0.95$

$\Rightarrow 0.05 = (1-e^{-\lambda t})^2 \Rightarrow 1-e^{-\lambda t} = \pm 0.2236 \Rightarrow e^{-\lambda t} = 0.7764, 1.2236$

$\Rightarrow \lambda = \frac{\ln 0.7764}{-1000}, \frac{\ln 1.2236}{-1000} = 0.000253, 0.000202.$

Hence, mean time to failure (MTTF) for component $= \int_0^\infty R(t)dt = \int_0^\infty e^{-\lambda t} dt = \frac{e^{-\lambda t}}{-\lambda}\Big|_0^\infty = \frac{1}{\lambda} = \frac{1}{0.000253} = 3951.$

" " " " for the system $= \int_0^\infty R_s(t)dt = \int [1-(1-e^{-\lambda t})^2]dt = \int_0^\infty (2e^{-\lambda t} - e^{-2\lambda t})dt$

$= \frac{2}{\lambda} - \frac{1}{2\lambda} = \frac{3}{2}\cdot\frac{1}{\lambda} = 5926.5$

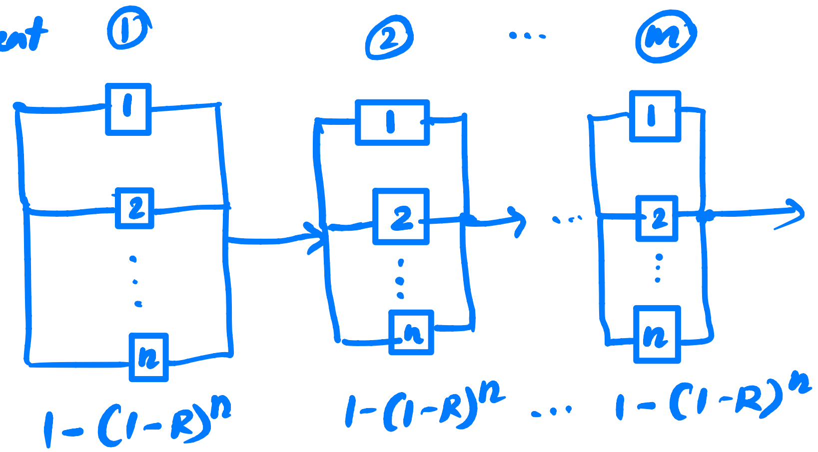
Parallel series configuration

- A system in which m subsystems are connected in series where each subsystem has n components connected in parallel. This is also called the low level redundancy.

Let the reliability of each component be $R(t)$.

Then, the reliability of each parallelly connected subsystem of n component has reliability $R_p(t) = 1-(1-R(t))^n$.

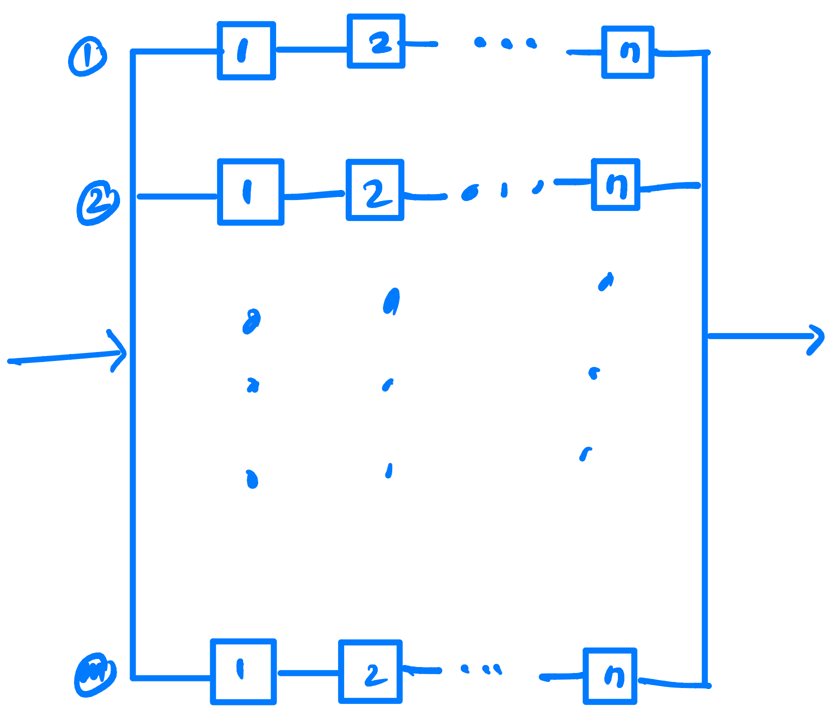
Finally, the system has reliability $R_s(t) = \left(1-(1-R(t))^n\right)^m$.



Series parallel configuration

- A system in which m subsystems are connected in parallel where each subsystem has n components connected in series. This is also called the high level redundancy.

Let each component has reliability $R(t)$. Then n-component connected in a series has reliability $= R(t)^n$.



Hence, the system has reliability

$$R_s(t) = 1 - (1-R(t)^n)(1-R(t)^n)\cdots(1-R(t)^n)$$
$$= 1-(1-R(t)^n)^m.$$